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1  %-----
2  % Regression (Octave)
3  %-----
4
5  clear
6  clc
7
8  pkg load statistics %For Octave
9
10 %Printing the outputs as a text file if ==1
11 output_file=0;
12 %LaTeX Interpreter if ==1 (requires LaTeX to be installed)
13 latex_interpreter=0;
14
15 %Sample size
16 n=500;
17
18 %True parameter values
19 beta_1=1;
20 beta_2=3;
21 beta_vec=[beta_1 beta_2]';
22
23 %Generating observations
24 rng(0); %Random number generator
25 x=randn(n,1); %(n,1)-vector of iid standard normal independent variables
26 e=randn(n,1); %(n,1)-vector of iid standard normal error terms
27 X=[ones(n,1) x];
28 y=X*beta_vec+e;
29
30 %Estimating best beta
31 beta_hat=inv(X'*X)*X'*y; %or: beta_hat=(X'*X)\(X'*y); Projection
32 beta_hat_2=regress(y,X); %MATLAB function regress
33
34 %Creating the regression line for Figure
35 x_min=-4;
36 x_max=4;
37 x_ax=linspace(x_min,x_max,100);
38
39 if output_file==1
40     dfile ='regression_Octave.txt';
41     if exist(dfile, 'file')
42         delete(dfile);
43     end
44     diary("regression_Octave.txt")
45     diary on
46 end
47
48 %Output (Command Window)
49 fprintf('Projection: beta=(%f,%f)\n',beta_hat(1),beta_hat(2));
50 fprintf('Matlab Function regress: beta=(%f,%f)\n',beta_hat(1),beta_hat(2));
51
52 if output_file==1
53     diary off
54 end
55
56 %Figure
57 figure()
58 hold on

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59 plot(x,y,'. '); %or: scatter(x,y);
60 plot(x_ax,beta_hat(1)+beta_hat(2)*x_ax);
61 grid on
62 hold off
63 xlim([x_min x_max]);
64 ylim([beta_1+beta_2*x_min beta_1+beta_2*x_max]);
65 if latex_interpreter==1
66     xlabel('$x$', 'Interpreter', 'latex', 'FontSize', 12);
67     ylabel('$y$', 'Interpreter', 'latex', 'FontSize', 12);
68     title('Regression: Example', 'Interpreter', 'latex', 'FontSize', 14);
69     %Putting the regression equation into the figure
70     reg_eq=sprintf('$y$ = %.3f + %.3f $x$', beta_hat(1), beta_hat(2));
71     text(-3.75, 2, reg_eq, 'Interpreter', 'latex', 'FontSize', 12);
72 else
73     xlabel('x', 'FontSize', 12);
74     ylabel('y', 'FontSize', 12);
75     title('Regression: Example', 'FontSize', 14);
76     %Putting the regression equation into the figure
77     reg_eq=sprintf('y = %.3f + %.3f x', beta_hat(1), beta_hat(2));
78     text(-3.75, 2, reg_eq, 'FontSize', 12);
79 end
80
81

```